Evaluating Size-Based Relational Reasoning in Vision–Language Models

Rotem Gatenyo (ID. 207984790), Ofek Basson (ID. 315682195)

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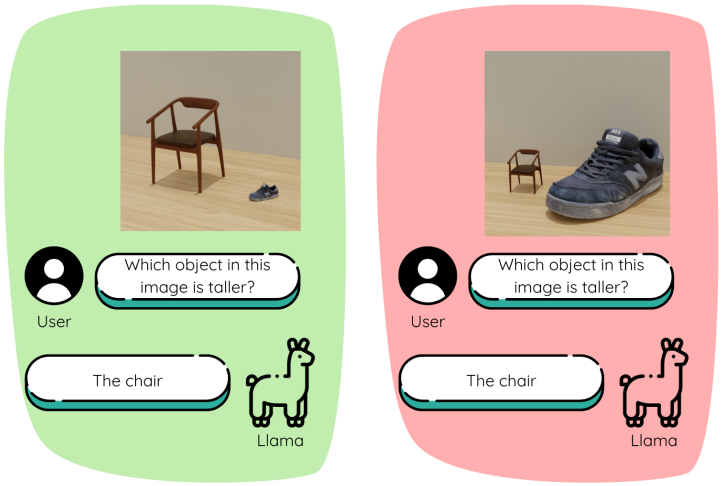


Figure 1: **Task illustration.** We pose the same question on two controlled renders of the same object pair. **Left**: the chair is taller than the shoe and the model answers "the chair" (correct; green background). **Right**: the shoe is rendered taller, yet the model again answers "the chair" (incorrect; red background). We apply this protocol across different images and object categories to test identification and taller-decision abilities.

1 Introduction

Recent advances in **Vision-Language Models (VLMs)** show strong multimodal abilities, from object recognition to relational reasoning. Yet, it remains unclear how these models handle **physical properties of objects**, especially when they deviate from real-world expectations.

In human perception, relative size conveys important cues (e.g., a chair is expected to be taller than a shoe). We therefore ask whether VLMs are sensitive

to such **size-based relations**, and whether altering object ratios affects their performance in identification and comparative reasoning.

To test this, we created controlled synthetic datasets in Blender that systematically vary object sizes, and queried the model with multiple formulations of identification and comparative questions to avoid wording bias.

Our goal is to assess whether changing relative size impairs a model's ability to identify objects and decide which is taller. While focusing on this main question, we also examine secondary factors such as the presence of object labels in the query.

This study sheds light on the **limitations and biases of VLMs** in grounded reasoning, and on how physical plausibility shapes their performance.

2 Related Work

2.1 Vision–Language Models (VLMs)

Early large VLMs combine strong language models with vision encoders to handle open-ended multimodal tasks such as *Visual Question Answering (VQA)*. *Flamingo* enables interleaved image - text inputs with strong few-shot performance [1]. *BLIP-2* introduces a lightweight connector to bridge frozen vision encoders and LLMs [2]. *PaLI/PaLI-3* scale joint multimodal modeling across tasks and languages [3]. *LLaVA* popularizes visually grounded, instruction-tuned chat [4]. In our experiments we use **Llama 3.2-Vision**, a recent multimodal variant in the Llama 3 family [5], as a representative modern VLM for our tests.

2.2 Attribute changes and counterfactual tests

Benchmarks that alter salient attributes probe whether models follow pixels or default to priors. *NEMO* pairs original and attribute-modified objects and reports sizable recognition gaps across models [6]. *ColorBench* evaluates color perception, reasoning, and robustness across diverse VLMs [7]. Complementarily, *On the Limitations of VLMs in Understanding Image Transforms* finds brittleness under simple transforms such as rotations and flips [8]. In our controlled renders, we *scale one object* so that the size ratio between the pair is intentionally *not natural*, yielding counterfactual yet fully controlled stimuli that specifically target *size* plausibility.

2.3 Spatial and relational reasoning

Recent work evaluates and improves spatial understanding. *SpatialVLM* assembles internet-scale spatial VQA supervision and improves both qualitative and *metric* spatial reasoning [9]. *SpaRE* builds a large *synthetic* spatial-reasoning VQA set and shows strong gains across spatial benchmarks [10]. Our "taller" comparison is a direct metric test of vertical extent within this family of spatial reasoning evaluations.

3 Methodology

3.1 Dataset Construction

We generated **10 distinct synthetic scenes** in Blender, each containing exactly two objects. To control for object variability, we selected items that typically have a **consistent real-world size**, such as books, shoes, cubes, cardboxes, fridges, wardrobes, etc. This ensured that judgments about scaling were less ambiguous.

For each scene, we rendered three image conditions:

1. **Normal** – both objects kept their natural proportions.
2. **Flipped** – the object that is originally smaller was scaled so that it appeared taller than the larger object.
3. **Even larger** – the object that is originally larger was scaled up even further, making the size difference more pronounced.

We used **height** (top-to-bottom distance) as the defining measurement, since it provides an **injective mapping** for size comparison and avoids ambiguity.

3.2 Question Design

To reduce the impact of question wording, we created **three categories of questions**, each with **five paraphrases**:

1. **Identification Ability** – asking the model to recognize and name both objects in the image.
2. **Taller-Decision (with object names)** – asking the model to compare the height of the two known objects.
3. **Taller-Decision (without object names)** – asking the model to decide which is taller without providing explicit labels.

This design allowed us to measure (i) **recognition performance**, and (ii) **reasoning about object height**, under both conditions where object names are provided and where they are not. Multiple paraphrasings ensured that conclusions were robust and not overly dependent on a single wording choice.

3.3 Evaluation Setup

For each image – question pair, the model's output was recorded and compared against the ground truth. Performance was measured across:

- **Accuracy** of object identification.
- **Accuracy** of height comparison.

By systematically varying object sizes and question formulations, we could directly assess whether **ratio manipulations affect VLM understanding**.

3.4 Human Study Design

Goal. Establish a human upper bound and verify that our synthetic renders are unambiguous for the two tasks we pose to the model: object identification and deciding which object is taller.

Stimuli. We reused the exact images from our dataset (Section 3.1), covering all categories and the three render conditions (normal, "flipped", and "even larger"). To keep exposure balanced, we sampled a fixed number of images per category and render condition.

Tasks. For each image, participants completed two prompts: (1) *Identification* - name the two objects (free text, short noun phrases accepted); (2) *Taller decision* - choose which object is taller (forced choice between the two objects).

Interface and procedure. Each participant saw *10* images (one per category). On each image, the two prompts were presented in randomized order. For the taller-decision prompt, the wording variant (with the object names shown vs. without) was randomized per image and balanced across the 10 images.

Participants and ethics. We recruited *32* adult participants.

Scoring. Identification responses were matched to ground truth with the same normalization and synonym rules used for the VLM (Section 3.3); taller responses were scored as correct/incorrect against the rendered height. We report per-image accuracy, then aggregate by category and by render condition.

4 Experimental results

4.1 VLM Study Results

We evaluated the vision - language model (VLM) on three tasks: (1) identifying objects correctly, (2) deciding which object is taller when object names were provided in the query, and (3) deciding which object is taller when object names were not provided. For each task, we compared performance across the three image conditions described in Section 3.1 (normal, "flipped", "even larger").

Accuracy was computed per scene as the proportion of correct responses. For each task, we compared the *normal* condition against the two altered conditions using paired Wilcoxon signed-rank tests. Two comparisons were performed per task, and Bonferroni correction was applied within each task.

For the task of **deciding which object is taller without being told the object names**, the VLM performed significantly better under the normal ratio compared to the flipped ratio ($p_{\text{adj}} = 0.0195$), with a mean accuracy advantage of +0.20 across 10 scenes. No difference was observed between the normal ratio and the "even larger" condition. For the task of **deciding which object is taller with object names given**, there was also an advantage for the normal ratio over the flipped ratio (mean +0.18), but although the raw p-value was relatively low ($p = 0.032$), this result did not remain significant after Bonferroni

Table 1: Wilcoxon signed-rank results summarized as mean accuracy *differences* (percentage points) of original-size images versus each altered condition. Positive values mean higher accuracy with original sizes. Adjusted p-values use Bonferroni

(a) Object identification

Condition (vs. original sizes)	**Mean diff**	**Bonferroni** p
Smaller object appears larger ("flipped")	+5.05	0.593
Larger object made even larger ("even larger")	+17.06	0.225
Summary	no significant change	

(b) Deciding which object is taller (with object names)

Condition (vs. original sizes)	**Mean diff**	**Bonferroni** p
Smaller object appears larger ("flipped")	+18.27	0.064
Larger object made even larger ("even larger")	−5.60	1.000
Summary	trend: original > flipped (n.s.)	

(c) Deciding which object is taller (without object names)

Condition (vs. original sizes)	**Mean diff**	**Bonferroni** p
Smaller object appears larger ("flipped")	+20.00	0.0195
Larger object made even larger ("even larger")	−2.60	1.000
Summary	original > flipped (significant)	

correction ($p_{\text{adj}} = 0.064$). Finally, in the **object identification task**, neither comparison was significant. There was a moderate mean advantage for normal over "even larger," but this effect was not statistically reliable ($p_{\text{adj}} = 0.225$). Results are presented in Table 1.

4.2 Human Study Results

Humans performed at ceiling on our tasks. Across all participants and categories, we collected **320 image trials** (32 participants × 10 images). For each image, participants answered both prompts, yielding **320 identification** responses and **320 taller-decision** responses.

These results confirm that our renders are unambiguous for people in both object identification and size comparison, including when one object is deliberately scaled to create an unnatural ratio. They also provide a clear human upper bound against which to interpret the VLM's performance. Results are presented in Table 2.

Table 2: Human accuracy (%) by render condition.

Condition	Identification	Taller (with names)	Taller (without names)
Normal	100	100	100
Flipped	100	100	100
Even Larger	100	100	100
All images	**100**	**100**	**100**

5 Discussion

5.1 VLM Study

Our analysis shows that manipulating relative object size had clear consequences for the model. In particular, when the smaller object was rendered larger than the larger one (flipped condition), the VLM's ability to decide which object was taller dropped significantly in the absence of object labels. This indicates that the model is sensitive to relative size distortions and struggles to recover correct comparisons when visual information alone conflicts with real-world size expectations. When object names were provided, the flipped manipulation also led to worse performance, although the effect was weaker and did not remain statistically significant after correction. This suggests that language cues can partially mitigate - but not fully eliminate - the disruptive effect of size inversions.

By contrast, exaggerating the size difference in the "even larger" condition had little effect on performance. This asymmetry highlights that the VLM is more vulnerable to violations of expected ratios than to exaggerations that preserve the correct ordering. Finally, object identification accuracy itself was not significantly impaired by either manipulation, though some comparisons yielded raw p-values close to the threshold. This implies that the model can usually recognize individual objects even when their relative sizes are distorted, but that this recognition does not necessarily translate into reliable comparative judgments.

Together, these results suggest that current VLMs may rely heavily on visual regularities for relational reasoning and only partly on textual labels as a compensatory signal. This points to a potential weakness in their generalization: they can label objects correctly but still fail in reasoning tasks when the visual scene violates learned priors. For practical applications, it emphasizes the need for training or evaluation regimes that explicitly probe robustness to visual manipulations that alter relational properties.

5.2 Human Study: Interpretation

Participants solved both tasks perfectly (**100%** accuracy) across all three render conditions - normal, "flipped", and "even larger". This establishes a human upper bound and confirms that the synthetic renders are unambiguous for both

object identification and the taller decision. Consequently, any errors or drops observed for the VLM - especially under counterfactual scaling or when object names are omitted - should be attributed to model limitations rather than dataset ambiguity or confusing prompts.

Unlike the model, humans showed no degradation when one object was deliberately scaled, indicating that people rely on the visible height in the image rather than real-world size priors. Second, providing object names had no measurable impact on human accuracy, suggesting that the images alone suffice for correct recognition and comparison.

6 Code Availability

All code and artifacts are available at `https://github.com/RotemGat/VLM-RelationalReasoning`.

The repository includes:

- VLM querying and data processing code.
- Evaluation pipeline that scores against GT and produces all tables/figures.
- Blender rendering assets.

References

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